

Computer Architecture and Organization
Mid-Term exam – Second Grade -15 Minutes

Time: 10:30 – 10:45

2023/2024 (Test 2-A)

فضلاً اجب في نفس الورقة - ممنوع استخدام الآلة الحاسبة

الاسم رباحي :

رقم المكتسب :

رقم المجموعة :

Questions

1. W

1. These types of interrupts can occur due to some illegal operation or the erroneous use of data.

- a. Trap Interrupts
- b. External Interrupts
- c. Internal Interrupts
- d. NO Interrupts

2.

2. In Assembly explain the program's purpose.

- a. Identifiers
- b. Directives
- c. Comments
- d. Labels

3.

3. In memory reference instructions, the address part is bits?

- a. 10
- b. 11
- c. 12
- d. 13

4

4. All of the following are reserved words in assembly language except

- a. SIZE
- b. FAR
- c. MODEL
- d. PAGE

5. The binary equivalent of $(37)_8$ is

- a. 11101111
- b. 01111000
- c. 00011111
- d. 11100111

6. The decimal equivalent of $(1101)_8$ is a. 577 b. 775 c. 575 d. 757	
7. Which of the following registers, keep track of the program instructions stored in memory? a. Accumulator b. Address Register c. Program Counter d. Index Register	
8. Which of the following is true about assembly language programs? a. They are machine-independent b. They are easier to read than high-level language programs c. They are faster and more efficient than high-level language programs d. They are harder to debug than high-level language programs	
9. which of the following instructions can have three operands a. MUL b. MOV. c. SHRD d. INC	
10. Which of the following is not a valid lable name a. start b. \$start c. __start d. !start	

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فضلاً أجب في نفس الورقة - ممنوع استخدام الآلة الحاسبة

..... : اعي

..... : ثن

..... : موعة

Questions	Answer
1. The binary equivalent of $(45)_{10}$ is a. 00100101 b. 01001010 c. 00101100 d. 10101010	
2. These types of interrupts generally come from external input / output devices a. Software Interrupts b. External Interrupts c. Internal Interrupts d. Trap Interrupts	
3. Inbus set-up is faster than other transmissions, so well suited for applications where messages are generated at irregular intervals a. Synchronous b. Semi-synchronous c. Asynchronous d. System	
4. is employed to hold temporarily the right-hand instruction from a word in memory. a. IR b. IBR c. MBR d. IMR	
5. Which of the following operations is/are performed by the ALU? a. Data manipulation b. Exponential c. Square root d. All of the above	

6. Which of the following instruction cannot be in octal program?	
a. 2008 b. 1010 c. 0652 d. 0053	
7. The decimal equivalent of $(247)_8$ is	
a. 167 b. 761 c. 617 d. 176	
8. Which of the following is not a typical assembly language instruction	
a. ADD b. MOV c. IF d. SHRD	
9.type indicates the boundary on which the segment is to begin.	
a. Alignment b. Combine c. Class d. Data	
10. In Signed Integers, if highest bit is 1 then it indicates that the number is	
a. Positive b. Negative c. does not matter d. invalid	

Wish You All the Luck with
Greetings

Course Instructors

Model test -6-

1. The _____ allows each individual interrupt to execute multiple functions.
A. Interrupt flag B. interrupt procedure ☒ C. interrupt macro D. interrupt subfunction
2. use both _____ and _____ instructions to read a character from the standard input.
A. Mov ah,02h, int 10h. ☒ B. Mov ah,01h, int 21h. C. Mov ah,0Eh, int 10h. D. Mov ah,09, int 21h
3. Before printing a string message using int 21h subfunction 09h the offset of the message should be stored in _____.
A. AX. ☒ B. BX. C. DX. D. CX.
4. After every response to the single step interrupt the flag that is cleared is.....
A. IF (Interrupt Flag) ☒ B. TF (Trap Flag) C. OF (Overflow Flag) ☒ D. None of the mentioned
5. The _____ contains an offset instead of actual address.
A. SP. ☒ B. IP. C. ES. ☒ D. SS.

1- For printing string on screen using int 21h we put In ah

a) 7

b) 8

~~c) 9~~

d) 10

2- Is that code true or false for reading characters from the keyboard without echo to the screen?

```
Mov ah,7
```

```
Int 10h
```

a) True

~~b) False~~

3- Consider a fully associative mapped cache of size 16 KB with block size 256 bytes. The size of the main memory is 128 KB.

1- Block offset

a) 10

b) 12

c) 14

~~d) 18~~

2- The physical address

a) 32

b) 34

~~c) 17~~

d) 36

3- The number of bits needed for the tag is.....

a) 8

~~b) 9~~

c) 7

d) 17

1.	The instruction 5 DUP(4) is equal to			
	a) 5,5,5,5	b) 4,4,4,4	c) 5,5,5,5,5	d) 4,4,4,4,4
2.	CMP instruction Subtract source operand from destination operand and store the result in destination operand			
	a) TRUE	b) FALSE		
3.	After Execute the following instructions al register contains and si register contains ... <pre> org 100h mov cl,5 mov si,0 mov al,0 x: add al, arr[si] inc si loop x ret arr db 1,2,3,4,5,6,7,8,9 </pre>			
	a) 10,5	b) 15,5	c) 5,10	d) 5,15
4.	PRINT_NUM procedure prints a signed number in			
	a) AX	b) BX	c) CX	d) DX

1- Using interrupt 10h for print character on screen we put in ah

- a) c b) d **c) e** d) f

10h $\left\{ \begin{array}{l} \rightarrow 0e \rightarrow \text{print (al)} \\ \rightarrow 02 \rightarrow \text{move cursor} \end{array} \right.$

2- For printing string on screen using int 21h we put in ah

- a) 7 b) 8 **c) 9** d) 10

21h $\left\{ \begin{array}{l} \rightarrow 02 \rightarrow \text{print (dl) char} \\ \rightarrow 09 \rightarrow \text{print string (al)} \end{array} \right.$

3- Consider a directly mapped cache of size 512 KB with a block size of 1 KB. There are 7 bits in the tag. Find-

1-Block offset

a) 10

b) 12

c) 14

d) 8

$$\begin{aligned} \text{Cache size} &= 512 \text{KB} \\ &= 2^9 \times 2^{10} \text{B} \end{aligned}$$

$$\text{Block size} = 1 \text{KB}$$

$$= 2^0 \times 2^{10} = 2^{10} \text{B}$$

$$\hookrightarrow \text{B.O} = 10 \text{ bit}$$

$$\text{Tag Bit} = 7 \text{ bit}$$

2-Cache index

a) 7

b) 9

c) 12

d) 10

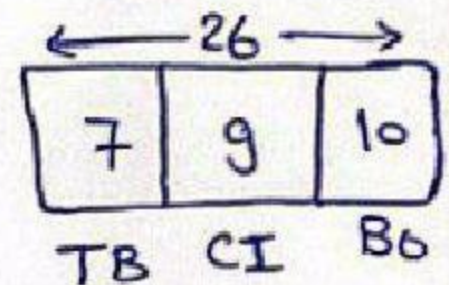
3-The physical address

a) 24

b) 25

c) 26

d) 28



$$\text{no. of line in cache} = \text{CS} / \text{BS}$$

$$= 2^{19} / 2^{10} = 2^9 \text{ line}$$

$$\text{C.I} = 9 \text{ bit}$$

$$\text{P.A} = 26 \text{ bit}$$

1. In IEEE double-precision representations, the significant part without sign bit occupies ____ bits.
(A) 8 (B) 23 (C) 52 (D) 11
2. Which cache mapping function(s) that may slow the access time?
(A) Direct (B) Direct & Full Associative (C) Set Associative (D) Full Associative